

Rapport TP 2 monitoring

Windows Server

Pour réaliser le TP je vais me servir de la version de l'ad en ligne de commande, et pour les installations, je vais le faire en ligne de commande sur powershell.

Téléchargement de windows explorer

```
PS C:\Users\Administrateur> Invoke-WebRequest -Uri "https://github.com/prometheus-community/windows_exporter/releases/download/v0.29.1/windows_exporter-0.29.1-amd64.msi" -OutFile "C:\windows_exporter.msi"
```

Ouverture du port car le pare-feu peut poser problème sur un Windows Server en version core.

```
PS C:\Users\Administrateur> New-NetFirewallRule -DisplayName "Windows Exporter" -Direction Inbound -LocalPort 9182 -Protocol TCP -Action Allow

Name
: {92237633-639d-4bc4-8fd8-205e96a3f65a}
DisplayName
: Windows Exporter
Description
:
DisplayGroup
:
Group
:
Enabled
: True
Profile
: Any
Platform
: {}
Direction
: Inbound
Action
: Allow
EdgeTraversalPolicy
: Block
LooseSourceMapping
: False
LocalOnlyMapping
: False
Owner
:
PrimaryStatus
: OK
Status
: La règle a été analysée à partir de la banque. (65536)
EnforcementStatus
: NotApplicable
PolicyStoreSource
: PersistentStore
PolicyStoreSourceType
: Local
RemoteDynamicKeywordAddresses
: {}
PolicyAppId
:
PackageFamilyName
:
```

On vérifie bien qu'on a accès aux métriques:

```
root@iut-snmip:~# curl http://192.168.255.100:9182/metrics
# HELP go_build_info Build information about the main Go module.
# TYPE go_build_info gauge
go_build_info{checksum="",path="github.com/prometheus-community/windows_exporter",version="(devel)"} 1
# HELP go_gc_duration_seconds A summary of the wall-time pause (stop-the-world) duration in garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 0
go_gc_duration_seconds{quantile="0.25"} 0
go_gc_duration_seconds{quantile="0.5"} 0
go_gc_duration_seconds{quantile="0.75"} 0
go_gc_duration_seconds{quantile="1"} 0.0011363
go_gc_duration_seconds_sum 0.0030557
go_gc_duration_seconds_count 19
# HELP go_gc_gogc_percent Heap size target percentage configured by the user, otherwise 100. This value is set by the GOGC environment variable, and the runtime/debug.SetGCPercent function. Sourced from /gc/gogc:percent
# TYPE go_gc_gogc_percent gauge
go_gc_gogc_percent 100
# HELP go_gc_gomemlimit_bytes Go runtime memory limit configured by the user, otherwise math.MaxInt64. This value is set by the GOMEMLIMIT environment variable, and the runtime/debug.SetMemoryLimit function. Sourced from /gc/gomemlimit:bytes
# TYPE go_gc_gomemlimit_bytes gauge
go_gc_gomemlimit_bytes 9.223372036854776e+18
```

```
< > ↺ Non sécurisé 192.168.255.100:9182/metrics
Note_20251001_10...

# HELP go_build_info Build information about the main Go module.
# TYPE go_build_info gauge
go_build_info{checksum="",path="github.com/prometheus-community/windows_exporter",version="(devel)"} 1
# HELP go_gc_duration_seconds A summary of the wall-time pause (stop-the-world) duration in garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 0
go_gc_duration_seconds{quantile="0.25"} 0
go_gc_duration_seconds{quantile="0.5"} 0
go_gc_duration_seconds{quantile="0.75"} 0
go_gc_duration_seconds{quantile="1"} 0.0011363
go_gc_duration_seconds_sum 0.0040966
go_gc_duration_seconds_count 21
```

L'ad est bien remonté dans prometheus:

Targets

ad (1/1 up) [show less](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://192.168.255.100:9182/metrics	UP	instance="192.168.255.100:9182" job="ad"	15.083s ago	194.9ms	

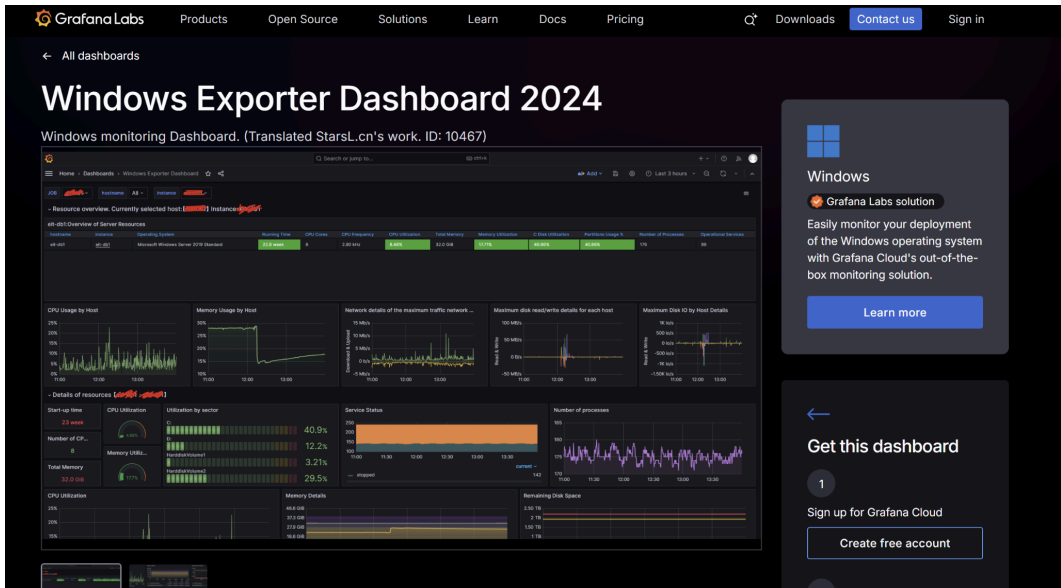
grafana (1/1 up) [show less](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:3000/metrics	UP	instance="localhost:3000" job="grafana"	8.543s ago	77.19ms	

Dans grafana aussi:

The screenshot shows the Grafana interface with the Prometheus data source selected. The 'Series limit' field is highlighted with a red box and contains the value '40000'. The 'window' label is also highlighted with a red box. The 'Go queryless' button is visible in the top right.

On va maintenant ajouter notre dashboard, dans un premier temps on recherche notre dashboard sur le site de grafana et on sélectionne celui qui nous intéresse et on copie le lien:



On colle le lien dans la partie encadrer (on peut aussi importer un fichier json directement):

Import dashboard

Import dashboard from file or Grafana.com



Upload dashboard JSON file

Drag and drop here or click to browse

Accepted file types: .json, .txt

Find and import dashboards for common applications at grafana.com/dashboards

Load

Import via dashboard JSON model

```
{
  "title": "Example - Repeating Dictionary variables",
  "uid": "_0HnEoN4z",
  "panels": [...]
  ...
}
```

LoadCancel

Il est impératif de vérifier que le dashboard fonctionne avec prometheus:

Import dashboard

Import dashboard from file or Grafana.com

Importing dashboard from [Grafana.com](#)

Published by	kakaru1331
Updated on	2024-05-30 09:08:26

Options

Name

Windows Exporter Dashboard 2024

Folder

Dashboards

Unique identifier (UID)

The unique identifier (UID) of a dashboard can be used for uniquely identify a dashboard between multiple Grafana installs. The UID allows having consistent URLs for accessing dashboards so changing the title of a dashboard will not break any bookmarked links to that dashboard.

Kdh0OoSGz

Change uid

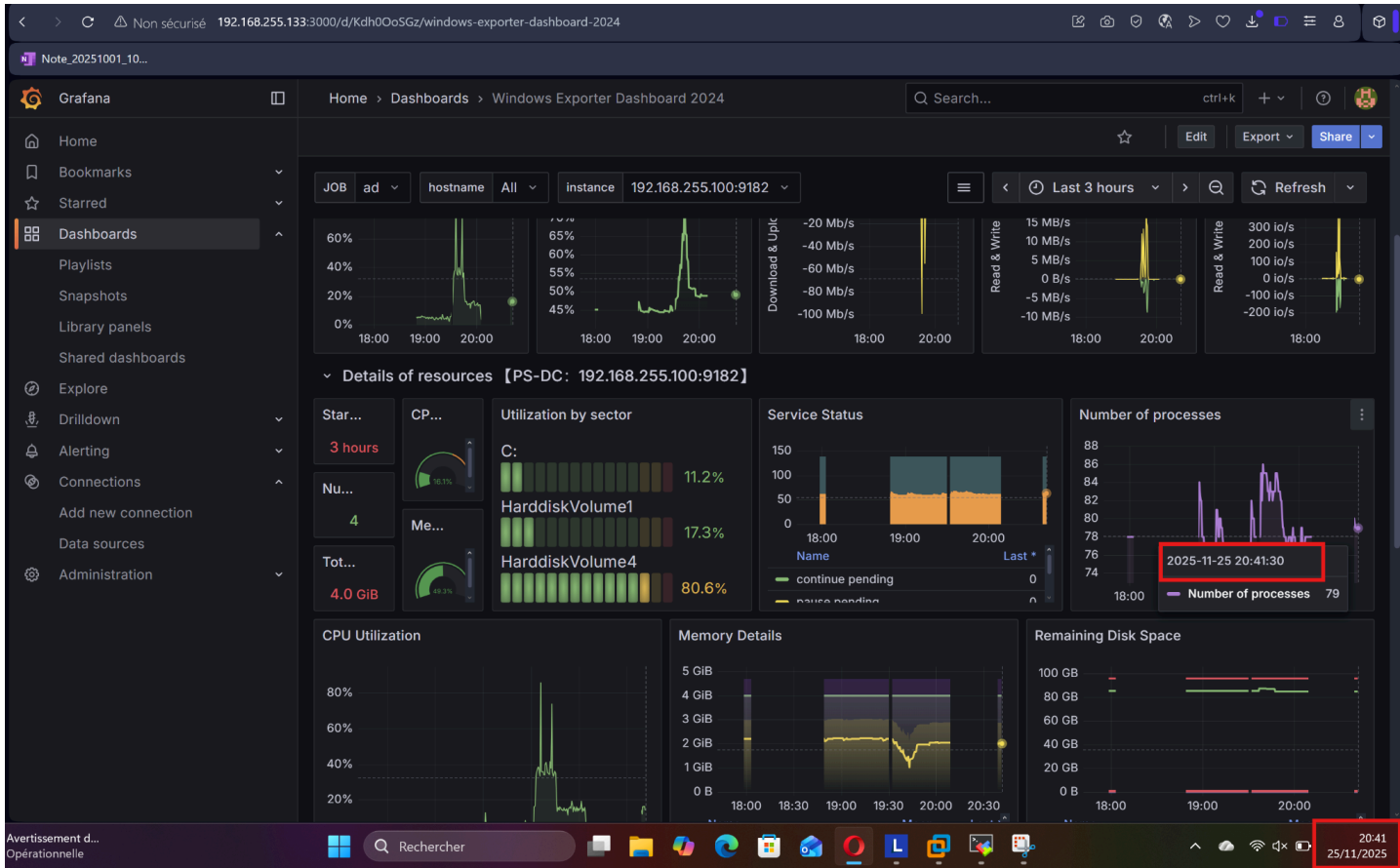
Prometheus

Prometheus

Import

Cancel

On se rend ensuite sur notre dashboard et on vérifie que tout marche bien:



Monitoring de Docker

On crée le fichier /etc/docker/daemon.json et on y met les lignes suivantes

```
GNU nano 8.4 daemon.json
{
  "metrics-addr" : "0.0.0.0:9323",
  "experimental" : true
}
```

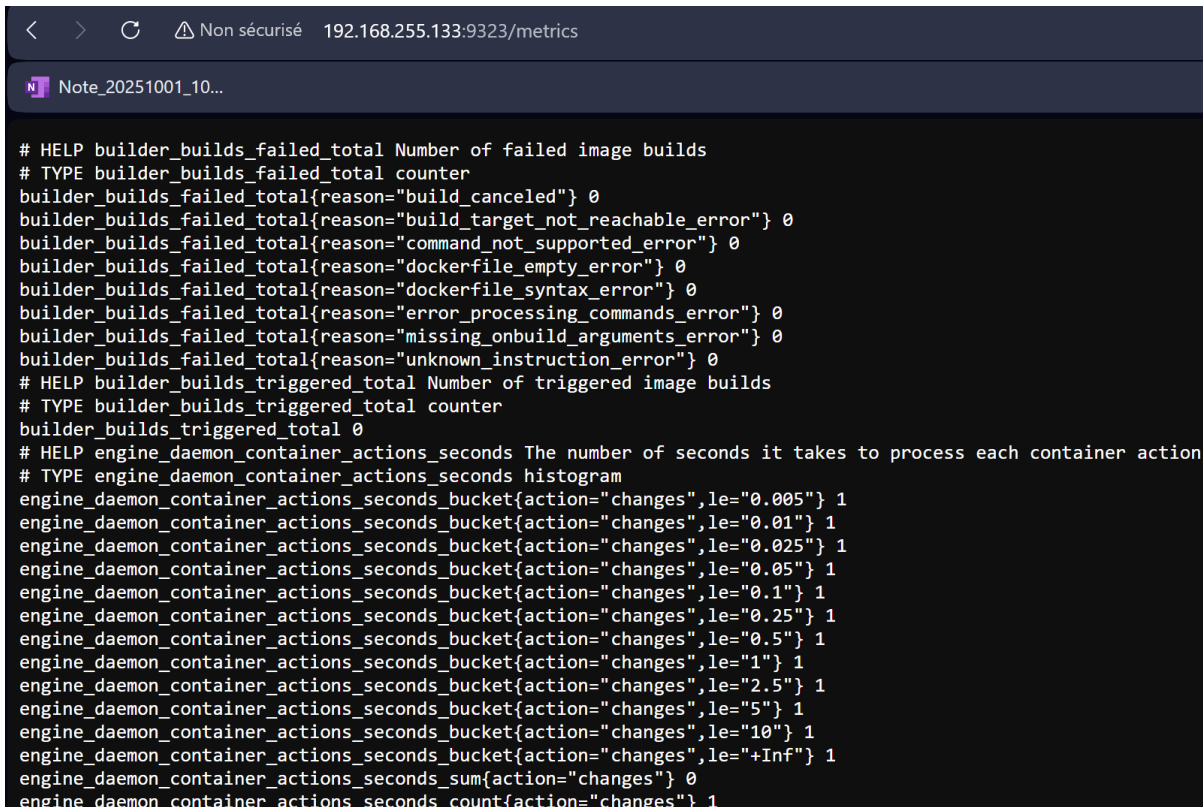
On redémarre docker:

```
root@iut-snm:/etc/docker# service docker restart
```

On voit bien le processus d'écoute

```
root@iut-snm:/etc/docker# ss -ltn
State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
LISTEN 0 128 0.0.0.0:22 0.0.0.0:* users:((("sshd",pid=997,fd=6))
LISTEN 0 128 127.0.0.1:6010 0.0.0.0:* users:((("sshd-session",pid=15345,fd=11))
LISTEN 0 4096 *:9323 *: users:((("dockerd",pid=18255,fd=17))
LISTEN 0 128 [::]:22 [::]:* users:((("sshd",pid=997,fd=7))
LISTEN 0 4096 *:9100 *: users:((("prometheus-node",pid=635,fd=3))
LISTEN 0 4096 *:9090 *: users:((("prometheus",pid=636,fd=6))
LISTEN 0 128 [::]:6010 [::]:* users:((("sshd-session",pid=15345,fd=10))
LISTEN 0 4096 *:3000 *: users:((("grafana",pid=2179,fd=11))
```

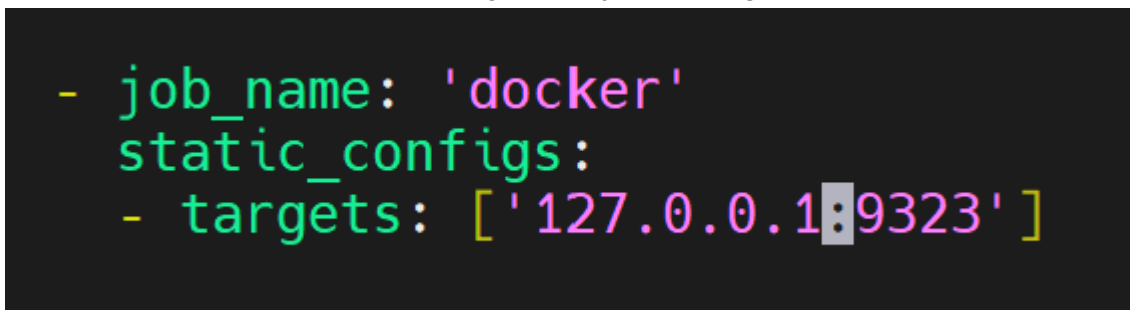
On vérifie qu'on accède bien aux métriques:



```
< > ↻ ⚠ Non sécurisé 192.168.255.133:9323/metrics
Note_20251001_10...

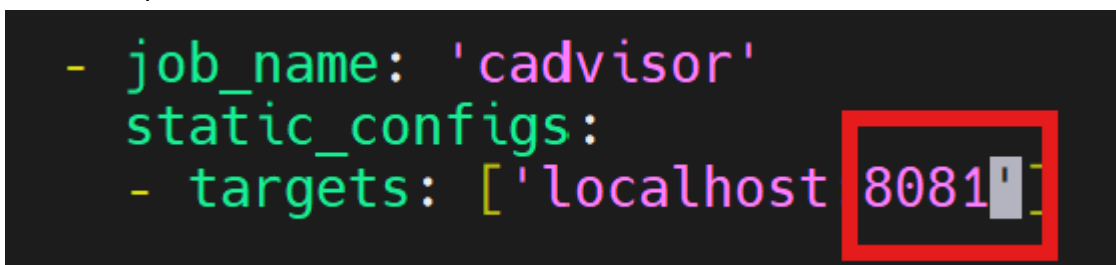
# HELP builder_builds_failed_total Number of failed image builds
# TYPE builder_builds_failed_total counter
builder_builds_failed_total{reason="build_canceled"} 0
builder_builds_failed_total{reason="build_target_not_reachable_error"} 0
builder_builds_failed_total{reason="command_not_supported_error"} 0
builder_builds_failed_total{reason="dockerfile_empty_error"} 0
builder_builds_failed_total{reason="dockerfile_syntax_error"} 0
builder_builds_failed_total{reason="error_processing_commands_error"} 0
builder_builds_failed_total{reason="missing_onbuild_arguments_error"} 0
builder_builds_failed_total{reason="unknown_instruction_error"} 0
# HELP builder_builds_triggered_total Number of triggered image builds
# TYPE builder_builds_triggered_total counter
builder_builds_triggered_total 0
# HELP engine_daemon_container_actions_seconds The number of seconds it takes to process each container action
# TYPE engine_daemon_container_actions_seconds histogram
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.005"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.01"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.025"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.05"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.1"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.25"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.5"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="1"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="2.5"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="5"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="10"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="+Inf"} 1
engine_daemon_container_actions_seconds_sum{action="changes"} 0
engine_daemon_container_actions_seconds_count{action="changes"} 1
```

Dans `/etc/prometheus/prometheus.yml` on ajoute ces lignes:



```
- job_name: 'docker'
  static_configs:
    - targets: ['127.0.0.1:9323']
```

Ensuite on ajoute aussi cadvisor dans `/etc/prometheus/prometheus.yml` en précisant le bon port et on redémarre prometheus:



```
- job_name: 'cadvisor'
  static_configs:
    - targets: ['localhost:8081']
```

cadvisor (1/1 up) show less					
Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:8081/metrics	UP	instance="localhost:8081" job="cadvisor"	11.656s ago	11.01 ms	

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All dashboards

cadvisor dashboard

I created this cadvisor dashboard to show everything scraped from cadvisor. - Expanding all the rows may consume too much resources. - Limit the containers by selecting compose_project. - First few rows are stacked. Container rows are not. If you have any issues or suggestions, please email mail@maoyin.eu.



←

Get this dashboard

1

Sign up for Grafana Cloud

Create free account

2

Import the dashboard template

Dashboard ID copied!

or

Download JSON

ID
19792

Datasource
Prometheus

Dependencies

On importe le dashboard:

Import dashboard

Import dashboard from file or Grafana.com



Upload dashboard JSON file

Drag and drop here or click to browse

Accepted file types: .json, .txt

Find and import dashboards for common applications at grafana.com/dashboards 

Load

Import via dashboard JSON model

```
{
  "title": "Example - Repeating Dictionary variables",
  "uid": "_0HnEoN4z",
  "panels": [...]
  ...
}
```

Load

Cancel

On sélectionne prometheus:

Import dashboard

Import dashboard from file or [Grafana.com](#)

Importing dashboard from [Grafana.com](#)

Published by	kokorinav
Updated on	2021-04-21 14:31:33

Options

Name

Cadvisor exporter

Folder

Dashboards

Unique identifier (UID)

The unique identifier (UID) of a dashboard can be used for uniquely identify a dashboard between multiple Grafana installs. The UID allows having consistent URLs for accessing dashboards so changing the title of a dashboard will not break any bookmarked links to that dashboard.

pMEd7m0Mz

Change uid

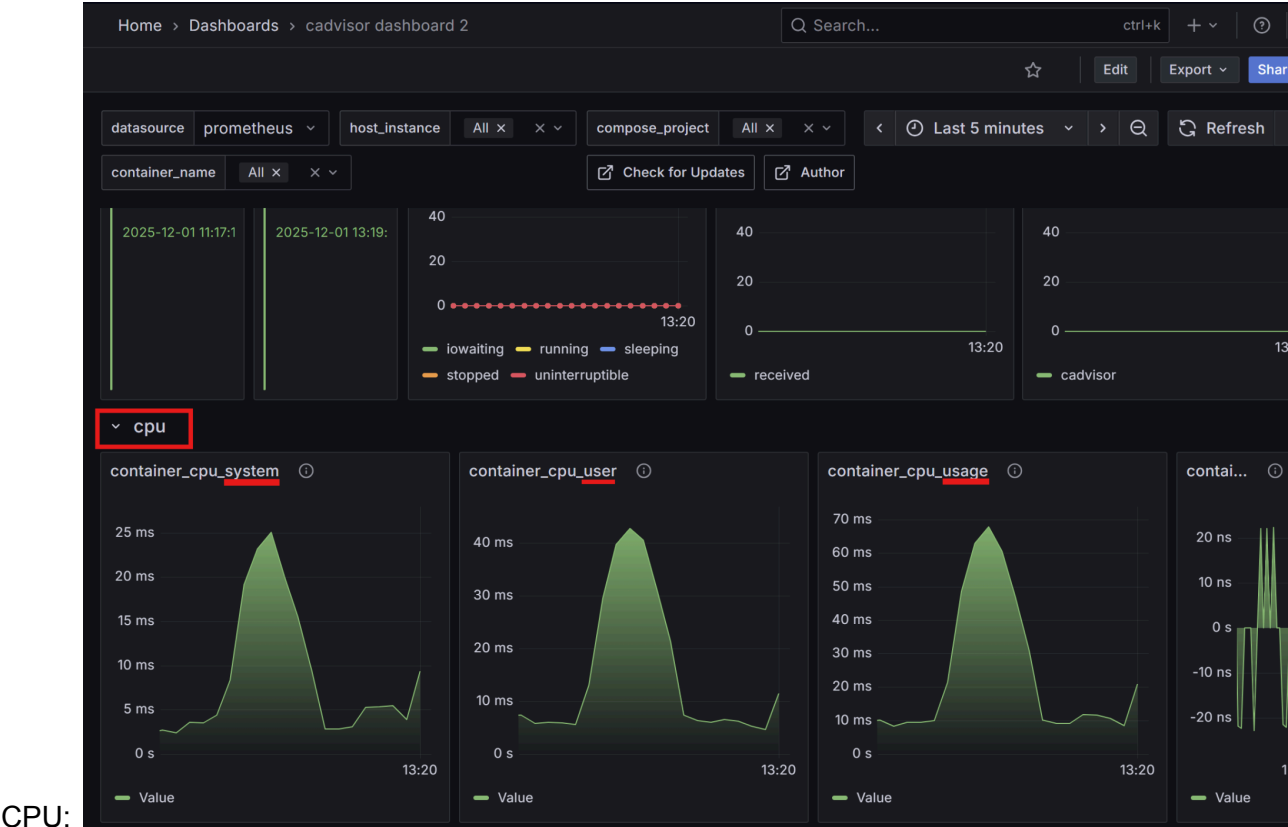
Prometheus

 prometheus

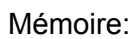
Import

Cancel

On vérifie qu'il récupère bien des données essentielle:



CPU:



On ajoute l'utilisateur et on download blackbox

```

root@tuit-snmpp:~# useradd --no-create-home --shell /bin/false blackbox_exporter
root@tuit-snmpp:~# wget https://github.com/prometheus/blackbox_exporter/releases/download/v0.25.0/blackbox_exporter-0.25.0.linux-amd64.tar.gz
--2025-12-01 13:51:40-- https://github.com/prometheus/blackbox_exporter/releases/download/v0.25.0/blackbox_exporter-0.25.0.linux-amd64.tar.gz
Résolution de github.com (github.com)... 140.82.121.3
Connexion à github.com (github.com) [140.82.121.3]:443... connecté.
requête HTTP transmise, en attente de la réponse... 302 Found
Emplacement : https://release-assets.githubusercontent.com/github-production-release-asset/41964498/86162a04-d27e-4276-8307-0031301d589c?sp=r&sv
=2018-11-09&sr=b&spr=https&size=2025-12-01T13%3A48%3A21Z&rscc=attachment%3Bfilename%3Dblackbox_exporter-0.25.0.linux-amd64.tar.gz&rscnt=applicatio
n%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2025-12-01T12%3A48%3A17Z&ske=2025-12-01T13%3A48%3A21Z&sk=s=bs&skv=2018-11-09&sig=YwLRbMrGvSwGcuJRv3CEk%2BH9JlDyBcX6js7r6JH4ATeM%3D&jwt=eYJ0eXA0IjIjV10iCjhbGciOiJIU2I1NiJ9.eYp3c3MiOiJnaXRodWU29tIiwiaXVxYXkiOiJmcmVzfWZfZS1hcnJlIHVud2UzOG93cy5uZXQ1fQ..FBEwBS2lPpAy-m6h1dIMu3cY9adHk3GpKvYUVX0i0&response-content-dispo
sition=attachment%3B%20filename%3Dblackbox_exporter-0.25.0.linux-amd64.tar.gz&rscnt=application%2Foctet-stream [suivait]
--2025-12-01 13:51:41-- https://release-assets.githubusercontent.com/github-production-release-asset/41964498/86162a04-d27e-4276-8307-0031301d5
89c?sp=r&sv=2018-11-09&sr=b&spr=https&size=2025-12-01T13%3A48%3A21Z&rscc=attachment%3Bfilename%3Dblackbox_exporter-0.25.0.linux-amd64.tar.gz&rscnt
=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2025-12-01T12%3A48%3A17Z&s
ke=2025-12-01T13%3A48%3A21Z&sk=s=bs&skv=2018-11-09&sig=YwLRbMrGvSwGcuJRv3CEk%2BH9JlDyBcX6js7r6JH4ATeM%3D&jwt=eYJ0eXA0IjIjV10iCjhbGciOiJIU2I1NiJ9.eYp3c3MiOiJnaXRodWU29tIiwiaXVxYXkiOiJmcmVzfWZfZS1hcnJlIHVud2UzOG93cy5uZXQ1fQ..FBEwBS2lPpAy-m6h1dIMu3cY9adHk3GpKvYUVX0i0&response-content-dispo
sition=attachment%3B%20filename%3Dblackbox_exporter-0.25.0.linux-amd64.tar.gz&rscnt=application%2Foctet-stream
Résolution de release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.110.133, 185.199.111.133, 185.199.108.133, ...
Connexion à release-assets.githubusercontent.com (release-assets.githubusercontent.com) [185.199.110.133]:443... connecté.
requête HTTP transmise, en attente de la réponse... 200 OK
Taille : 11269099 (11M) [application/octet-stream]
Sauvegarde en « blackbox_exporter-0.25.0.linux-amd64.tar.gz »

blackbox_exporter-0.25.0.linux-amd64 100%[=====] 10,75M 11,5MB/s ds 0,9s

2025-12-01 13:51:42 (11.5 MB/s) - « blackbox_exporter-0.25.0.linux-amd64.tar.gz » sauvegardé [11269099/11269099]

```

On crée le fichier de conf de blackbox:

```
GNU nano 8.4 /etc/systemd/system/blackbox_exporter.service
[Unit]
Description=Blackbox Exporter
Wants=network-online.target
After=network-online.target

[Service]
User=blackbox_exporter
Group=blackbox_exporter
Type=simple
ExecStart=/etc/blackbox/blackbox_exporter --config.file /etc/blackbox/blackbox.yml

[Install]
WantedBy=multi-user.target
```

On lance le service et on s’assure qu’il se lancera à chaque fois:

```
root@iut-snmp:~# systemctl daemon-reload
root@iut-snmp:~# systemctl start blackbox_exporter
root@iut-snmp:~# systemctl enable blackbox_exporter
Created symlink '/etc/systemd/system/multi-user.target.wants/blackbox_exporter.service' → '/etc/systemd/system/blackbox_exporter.service'.
root@iut-snmp:~# systemctl status blackbox_exporter
● blackbox_exporter.service - Blackbox Exporter
   Loaded: loaded (/etc/systemd/system/blackbox_exporter.service; enabled; preset: enabled)
   Active: active (running) since Mon 2025-12-01 14:09:51 CET; 19s ago
 Invocation: 8b17f4ad2c4d4ad591afe8356bedf095
   Main PID: 72981 (blackbox_export)
    Tasks: 6 (limit: 2272)
   Memory: 4.7M (peak: 5.1M)
      CPU: 47ms
   CGroup: /system.slice/blackbox_exporter.service
           └─72981 /etc/blackbox/blackbox_exporter --config.file /etc/blackbox/blackbox.yml
```

Blackbox est donc opérationnelle sur le port 9115:

< > ↺ ⚠ Non sécurisé 192.168.255.133:9115

N Note_20251001_10...

Blackbox Exporter

[Probe prometheus.io for http_2xx](#)
[Debug_probe prometheus.io for http_2xx](#)
[Metrics](#)
[Configuration](#)

Recent Probes

Module	Target	Result	Debug
--------	--------	--------	-------

On ajoute le code suivant dans /etc/prometheus/prometheus.yml et on redémarre notre service :

```
#BLOC BLACKBOX
- job_name: "blackbox-http"
  metrics_path: "/probe"
  params:
    module: [ "http_2xx" ]
  relabel_configs:
    - source_labels: [ "__address__" ]
      target_label: "__param_target"
    - source_labels: [ "__param_target" ]
      target_label: "instance"
    - target_label: "__address__"
      replacement: "localhost:9115"
  static_configs:
    - targets: [ "https://prometheus.io/", "https://grafana.com/" ]
```

Tout apparaît bien sur prometheus:

Prometheus Alerts Graph Status Help New UI

All Unhealthy Collapse All

ad (0/1 up) show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://192.168.255.100:9182/metrics	DOWN	instance="192.168.255.100:9182/" job="ad"	12.584s ago	3.079s	Get "http://192.168.255.100:9182/metrics": dial tcp 192.168.255.100:9182: connect: no route to host

blackbox-http (2/2 up) show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:9115/probe module="http_2xx" target="https://grafana.com/"	UP	instance="https://grafana.com/" job="blackbox-http"	6.74s ago	611.6ms	
http://localhost:9115/probe module="http_2xx" target="https://prometheus.io/"	UP	instance="https://prometheus.io/" job="blackbox-http"	14.073s ago	424.2ms	

cadvisor (1/1 up) show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
----------	-------	--------	-------------	-----------------	-------

On importe notre dashboard:

Import dashboard

Import dashboard from file or Grafana.com



Upload dashboard JSON file

Drag and drop here or click to browse

Accepted file types: .json, .txt

Find and import dashboards for common applications at grafana.com/dashboards

Load

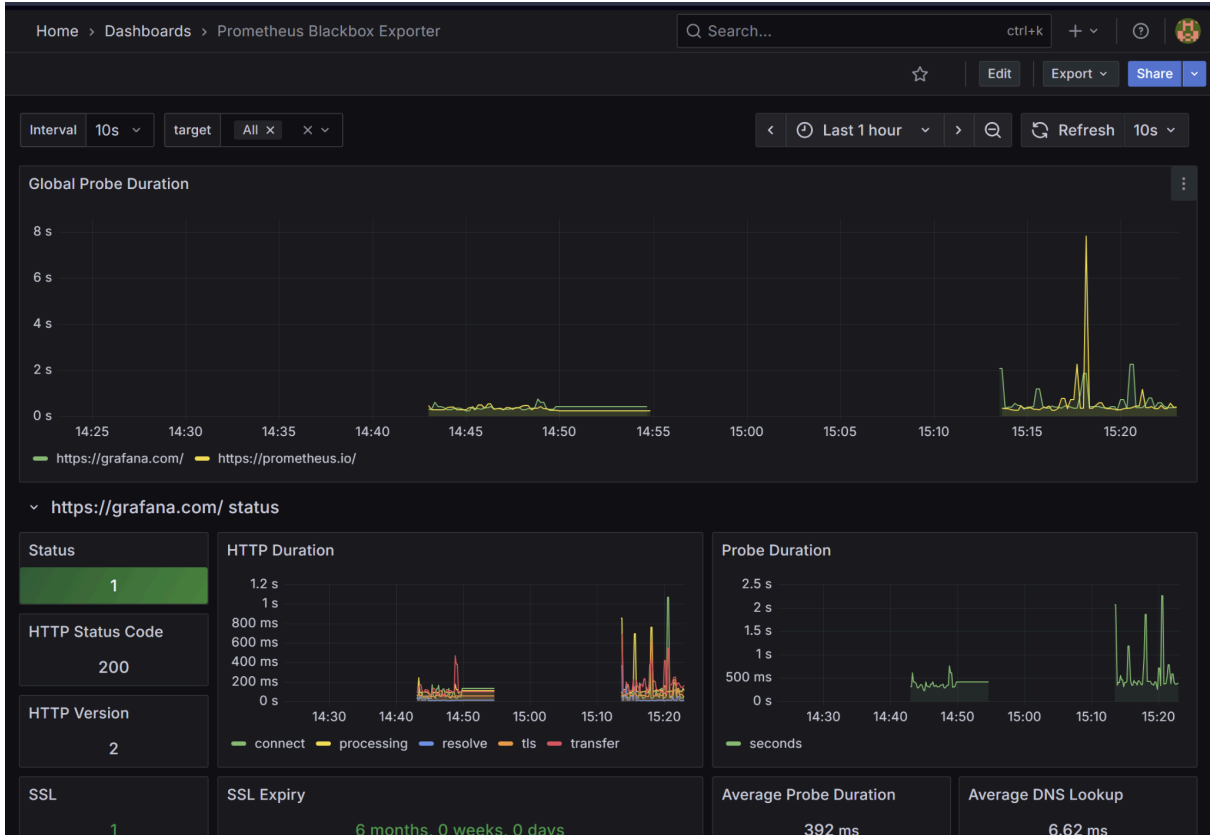
Import via dashboard JSON model

```
{
  "title": "Example - Repeating Dictionary variables",
  "uid": "_0HnEoN4z",
  "panels": [...],
  ...
}
```

Load

Cancel

Le dashboard est fonctionnelle:



Pour lier notre wordpress a notre dashboard on va effectuer une modification sur le fichier de conf de prometheus et on relance notre prometheus:

```
GNU nano 8.4 /etc/prometheus/prometheus.yml *
# metrics_path defaults to '/metrics'
# scheme defaults to 'http'.

static_configs:
- targets: ['localhost:9090']

- job_name: 'grafana'
  static_configs:
  - targets: ['localhost:3000']
  # If prometheus-node-exporter is installed, grab stats about the local
  # machine by default.
- job_name: 'node'
  static_configs:
  - targets: ['localhost:9100']

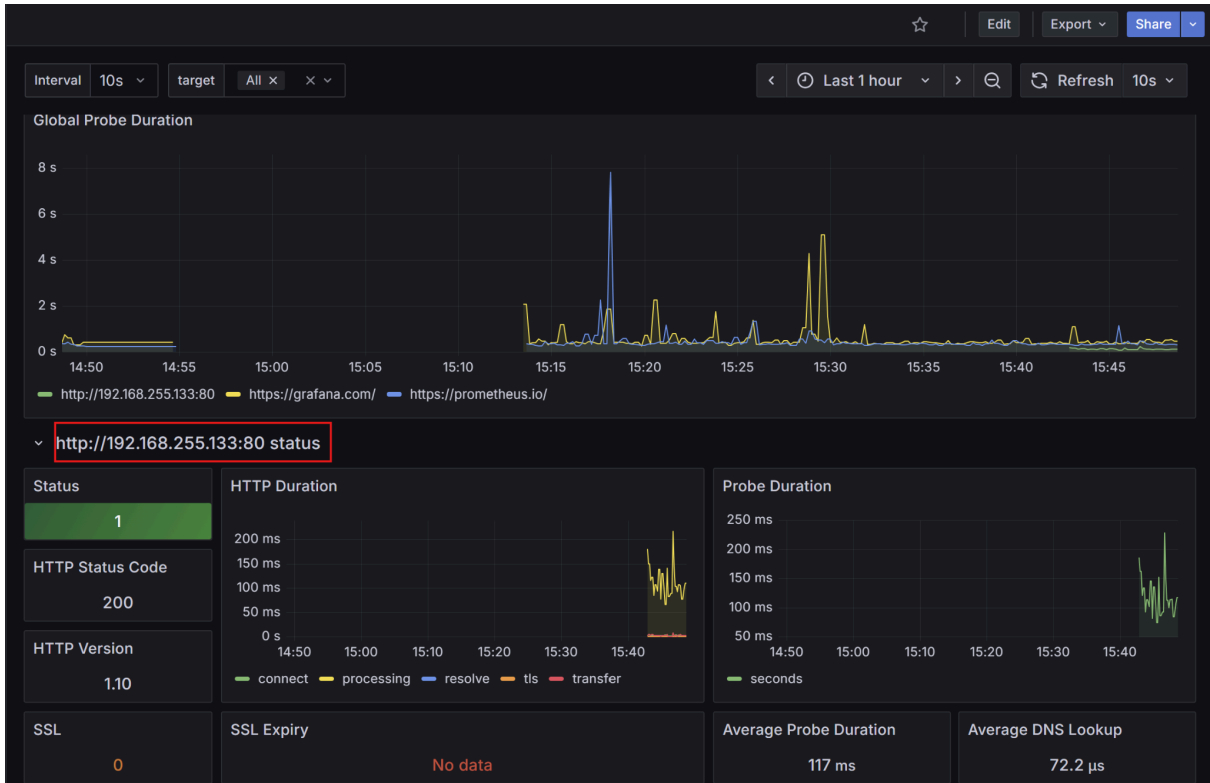
- job_name: 'ad'
  static_configs:
  - targets: ['192.168.255.100:9182']

- job_name: 'docker'
  static_configs:
  - targets: ['127.0.0.1:9323']

- job_name: 'cadvisor'
  static_configs:
  - targets: ['localhost:8081']

#BLOC BLACKBOX
- job_name: "blackbox-http"
  metrics_path: "/probe"
  params:
    module: [ "http_2xx" ]
  relabel_configs:
  - source_labels: [ "__address__" ]
    target_label: "__param_target"
  - source_labels: [ "__param_target" ]
    target_label: "instance"
  - target_label: "__address__"
    replacement: "localhost:9115"
  static_configs:
  - targets: [ "https://prometheus.io/", "https://grafana.com/", 'http://192.168.255.133:80' ]
```

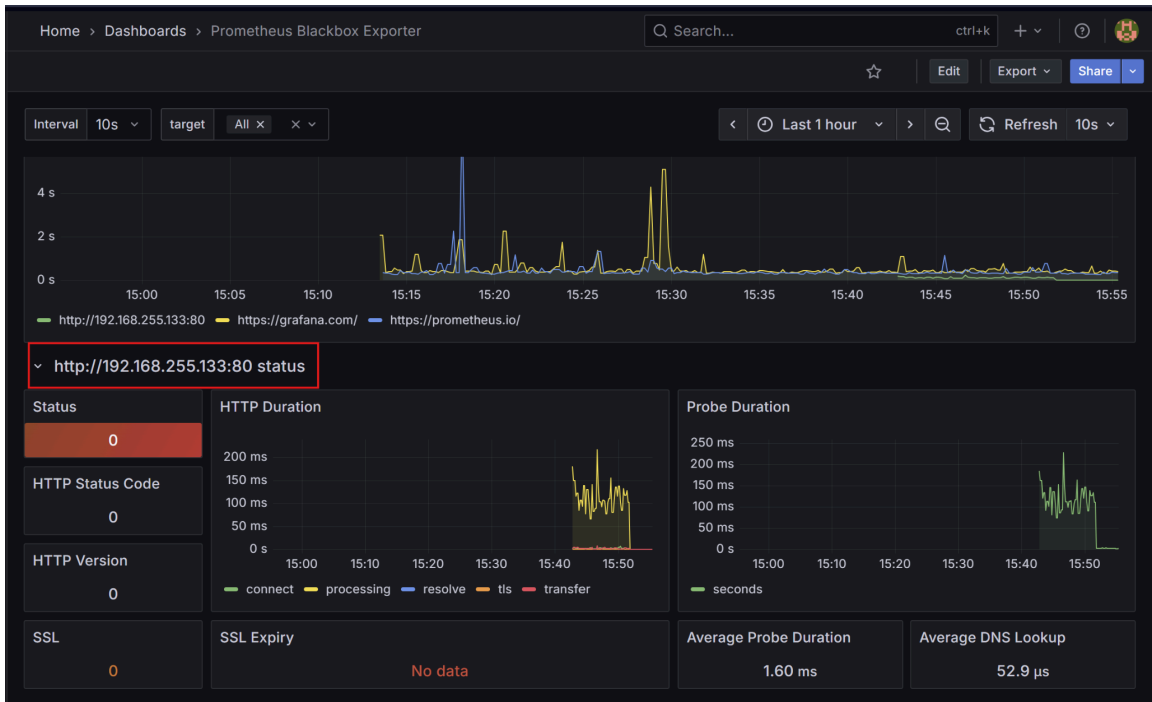
Notre wordpress a bien été ajouté à notre interface il est bien fonctionnelle:



On va stopper notre docker wordpress pour s'assurer que le dashboard soit bien mis à jour:

```
root@iut-snmp:~# docker stop wordpress
wordpress
```

Notre graph c'est bien mis à jour:



Configuration de l'alerte mail

On commence par installer alertmanager:

[illegible]

Installation manuelle des binaires d'Alertmanager et initialisation de sa configuration:

```
root@iut-snmp:/etc/prometheus# tar xvf alertmanager-0.26.0.linux-amd64.tar.gz
alertmanager-0.26.0.linux-amd64/
alertmanager-0.26.0.linux-amd64/alertmanager.yml
alertmanager-0.26.0.linux-amd64/NOTICE
alertmanager-0.26.0.linux-amd64/amtool
alertmanager-0.26.0.linux-amd64/alertmanager
alertmanager-0.26.0.linux-amd64/LICENSE
root@iut-snmp:/etc/prometheus# mv alertmanager-0.26.0.linux-amd64/alertmanager /usr/local/bin/
root@iut-snmp:/etc/prometheus# mkdir /etc/alertmanager
root@iut-snmp:/etc/prometheus# nano /etc/alertmanager/alertmanager.yml
```

On crée un mot de passe application sur google:

← Mots de passe des applications

Les mots de passe d'application vous permettent de vous connecter à votre compte Google sur des applis et des services plus anciens, non compatibles avec les normes de sécurité les plus récentes.

Les mots de passe d'application sont moins sécurisés que les applis et services à jour qui utilisent les normes de sécurité les plus récentes. Avant de créer un mot de passe d'application, vous devez vérifier si votre appli en a besoin pour établir la connexion.

[En savoir plus](#)

Vos mots de passe d'application

BlackBoxMonitoring

Date de création : 10:25



On fait notre conf en précisant le serveur de google, notre mail et le mdp d'application qu'on a créé juste avant:

```
GNU nano 8.4 /etc/alertmanager/alertmanager.yml *
global:
  resolve_timeout: 5m
  # Configuration du serveur Gmail
  smtp_smarthost: 'smtp.gmail.com:587'
  smtp_from: 'paulin.kuhn05@gmail.com'
  smtp_auth_username: 'paulin.kuhn05@gmail.com'
  smtp_auth_password: 'ogbm isrw kwpz wbon' # Les 16 lettres (sans espaces si possible)
  smtp_require_tls: true

route:
  group_by: ['alertname']
  group_wait: 10s
  group_interval: 10s
  repeat_interval: 1h
  receiver: 'email-moi'

receivers:
- name: 'email-moi'
  email_configs:
  - to: 'paulin.kuhn05@gmail.com' # Là où tu veux recevoir l'alerte
    send_resolved: true # Envoie un 2ème mail quand le problème est réglé
```

On crée le fichier de service pour gérer Alertmanager. Cela permet de le lancer automatiquement au démarrage et de le faire tourner en arrière-plan sans bloquer le terminal:

```
root@iut-snmp:~# nano /etc/systemd/system/alertmanager.service
```

```

GNU nano 8.4 /etc/systemd/system/alertmanager.service
[Unit]
Description=Alertmanager Service
Wants=network-online.target
After=network.target

[Service]
User=alertmanager
Group=alertmanager
Type=simple
ExecStart=/usr/local/bin/alertmanager --config.file=/etc/alertmanager/alertmanager.yml
Restart=always

[Install]
WantedBy=multi-user.target

```

On crée un fichier rules.yml pour définir nos conditions d'alerte (ici, quand un site ne répond plus):

```

GNU nano 8.4 /etc/prometheus/rules.yml *
groups:
- name: mes_alertes_sites
  rules:
  - alert: SiteWebDown
    expr: probe_success == 0
    for: 1m
    labels:
      severity: critical
    annotations:
      summary: "Le site {{ $labels.instance }} ne répond plus !"

```

On l'ajoute dans la configuration principale de Prometheus pour qu'il prenne en compte cette nouvelle règle.

```

# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
- "/etc/prometheus/rules.yml"

```

On redémarre alerte manager et on l'active au démarrage de la vm:

```

systemctl daemon-reload
# Démarrer Alertmanager
systemctl start alertmanager
# L'activer au démarrage
systemctl enable alertmanager
# Vérifier le statut
systemctl status alertmanager
Created symlink '/etc/systemd/system/multi-user.target.wants/alertmanager.service' -> '/etc/systemd/system/alertmanager.service'.
● alertmanager.service - Alertmanager Service
   Loaded: loaded (/etc/systemd/system/alertmanager.service; enabled; preset: enabled)
   Active: activating (auto-restart) (Result: exit-code) since Tue 2025-12-02 13:48:20 CET; 42ms ago
 Invocation: 1e52d91d7780470fbdc7c02a93ecc52e
   Process: 9154 ExecStart=/usr/local/bin/alertmanager --config.file=/etc/alertmanager/alertmanager.yml (code=exited, status=1/FAILURE)
   Main PID: 9154 (code=exited, status=1/FAILURE)
  Mem peak: 29.9M
    CPU: 106ms

```

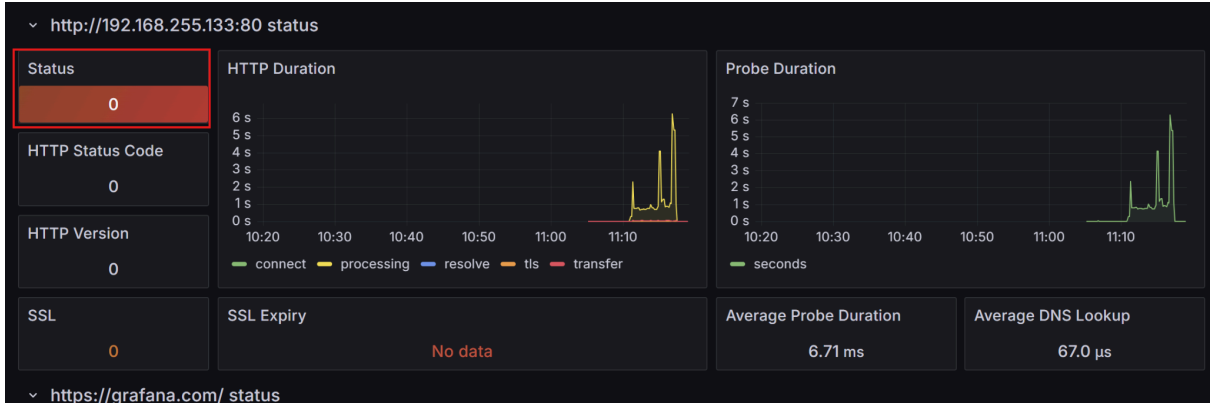
On va maintenant tester si notre service fonctionne bien, on désactive notre docker wordpress :

```

root@iut-snm:~# docker stop wordpress
wordpress

```

Notre service est bien couper:



L’alerte est bien déclenché dans prometheus:

Alerts

Inactive (0) Pending (1) Firing (0)

Show annotations

/etc/prometheus/rules.yml > mes_alertes_sites

SiteWebDown (1 active)

```
alert: SiteWebDown
expr: probe_success
== 0
for: 1m
labels:
  severity: critical
annotations:
  summary: Le site {{ $labels.instance }} ne répond plus !
```

Labels	State	Active Since	Value
alertname="SiteWebDown" instance="http://192.168.255.133:80" job="blackbox-http" severity="critical"	PENDING	2025-12-03 10:34:11.006916664 +0000 UTC	0

Et on reçoit bien le mail:

[FIRING:1] SiteWebDown (http://192.168.255.133:80 blackbox-http example critical) Boîte de réception x

Résumer cet e-mail

Paulin Kuhn paulin.kuhn05@gmail.com 11:35 (il y a 1 minute)

1 alert for alertname=SiteWebDown

View In Alertmanager

[1] Firing

Labels

alertname = SiteWebDown

instance = <http://192.168.255.133:80>

job = blackbox-http

monitor = example

severity = critical

Annotations

summary = Le site <http://192.168.255.133:80> ne répond plus !

[Source](#)